Muhammad Ameer Hamza

Full-Stack Engineer · Next.js · TypeScript · Python · PostgreSQL

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Portfolio · LinkedIn · GitHub

Summary

Full-stack engineer who builds business platforms end to end, from requirements to production support. Sole developer of five business platforms and two automation suites for finance, accounting, hospitality and pharma distribution, including a reporting platform used across 20–25 companies that cut each weekly report from up to an hour to about 5 minutes. Works remotely with a UK firm across London and Pakistan timezones, and writes extensively tested code (200–500+ automated tests on each main platform).

Experience

Full-Stack Software Engineer · MCA Global (Main Course Associates), UK hospitality finance firm · Jul 2024 – Present · Islamabad / Remote with London HQ

MCA Orbit: budgeting and weekly flash-reporting platform (mcaorbit.com), sole developer

- Built end to end, from requirements to deployment and support, an internal multi-tenant platform (Next.js, TypeScript, Python FastAPI, PostgreSQL) used by 15–20 finance staff and client users across 20–25 hospitality companies.
- Automated weekly flash P&L reports with written commentary, cutting each report from 30–60 minutes of spreadsheet work to about 5 minutes (83–92% faster) and removing manual calendar errors that skewed budgets.
- Built recency-weighted budget generation from multi-year history with growth targets and UK holiday seasonality, plus inter-site cost allocation rules with cycle checks.
- Cut the company dashboard from 130 queries / 19.5 s to 10 queries / 2.2 s (~9× faster); added company-scoped authorization on every route, revocable sessions and rate limiting; 219 automated tests.

MCA Pro: practice-management platform (mcapro.co), sole developer

- Built an internal practice-management platform (Next.js 16, TypeScript, PostgreSQL/Prisma) for teams in London and Pakistan, managing clients' recurring VAT, CT600, accounts and payroll work.
- Encoded UK statutory deadlines as code and automated daily reminder emails across two timezones with overdue escalation, using duplicate-proof scheduling enforced by Postgres constraints.
- Implemented role-based access for 5 roles across 2 branches, TOTP two-step verification with AES-256-GCM encrypted secrets, preparer/reviewer sign-off and a full audit trail; ~526 automated tests.

mcaaudit: AI accounts-review platform, sole developer

- Designed and built an AI accounts-review platform (Python/FastAPI, React 19/TypeScript, Claude API) that automates the first-pass review of ledgers, bank statements and management accounts packs, which took accountants 30 minutes for clean books and 2–3 hours when errors had to be found.
- Built a parser for Excel, CSV, PDF statements and accounting-software exports, plus 19 automated checks including a three-pass bank-to-books reconciliation, catching duplicates, unbalanced vouchers, balance breaks and transposition errors down to the exact row or cell.
- Kept the maths in code and the judgement in the AI: every row the AI cites is verified against the file, junior staff are never shown fixes (three-layer safeguard), and admin feedback builds a training library; prompt caching and per-user budget caps keep costs predictable.

Automation

- Built a Python + Playwright RPA suite of 5 bots (13 deployments across 11 client sites) that log into supplier and utility portals (Booker, FreshDirect, BT, Albion, Castle Water), download new invoices and email them to bookkeeping; 950+ invoices delivered, each exactly once, through crash-safe state tracking, verified Gmail API sends and automatic retry of pending invoices.
- Built a Python + Gmail API email automation that reads supplier emails and their PDFs, classifies each as invoice, statement or query, and forwards only invoices to AutoEntry (Sage); deployed across 16 sites for 9 hospitality clients, monitoring ~460 supplier senders, running unattended since Jul 2025 (1,600+ invoices forwarded at one site alone).

Company website

- Redesigned and rebuilt the firm's client-facing website (PHP, vanilla JS) with a custom flat-file CMS, scroll-driven animation and interactive finance tools (deal-value builder, gross-profit calculator, funding quiz); cut image weight 57% and added bot-resistant lead capture.

Founder & Full-Stack Developer · Freelance and own products · 2025 – Present · Remote

RestoCore: restaurant POS SaaS for Pakistan (own product, in pilot)

- Built a multi-tenant restaurant POS and back office (Next.js, TypeScript, Supabase/PostgreSQL): till, live kitchen display, inventory and recipe costing, payroll and reporting, with 4 subscription plans and row-level-security tenant isolation.
- Built electronic tax invoicing for PRA, KPRA and FBR with one adapter per authority and a retry queue; invoices are never marked valid unless the authority accepts them.
- Designed an offline till with exactly-once upload and database-level locking for kitchen and payment concurrency; cut send-to-kitchen time from 6.5 s to 3.5 s; ~274 automated tests.

Pharma distribution system: freelance build for a medicine distributor (product owned and licensable)

- Built a custom billing, stock and credit-ledger system (Next.js 15, TypeScript, PostgreSQL/Prisma) that replaced the client's previous system, fixing their pricing, stock and recovery problems; 49 screens and 87 API endpoints.
- Engineered exactly-once offline sync (service worker, IndexedDB outbox, idempotency keys, reserved invoice numbers) so billing continues through internet outages with no duplicate or lost bills.
- Designed concurrency-safe expiry-first (FEFO) stock allocation and paisa-accurate weighted-average costing using row and advisory locks, validated with 100-way concurrency benchmarks.

fbreinvoicing.com: FBR e-invoicing SaaS

- Built and launched a multi-tenant e-invoicing SaaS integrated with FBR's API (Next.js, NestJS, PostgreSQL, Docker on AWS); closed it in 2026 when leads didn't convert, and reused the integration experience in RestoCore.

Web Developer Intern · National Information Technology Board (NITB) · Jul – Aug 2024 · Islamabad

- Worked on internal government dashboards and built backend APIs with MongoDB.

Skills

- **Languages:** TypeScript, JavaScript, Python, PHP, SQL
- **Frontend:** Next.js, React, Tailwind CSS, React Query, Zustand, shadcn/ui
- **Backend:** Node.js, Next.js Route Handlers & Server Actions, FastAPI, NestJS, REST APIs
- **Data:** PostgreSQL (row-level security, locking, sequences), Prisma, Supabase, Neon, MongoDB
- **Cloud & DevOps:** Vercel, Render, AWS EC2, Docker, Nginx, cron jobs
- **Automation & AI:** Python RPA (Playwright), Gmail API automation, PDF parsing, Claude and OpenAI APIs (prompt caching, retrieval)
- **Testing & security:** Playwright, Vitest, Jest; RBAC, TOTP 2FA, encryption, audit logging, offline-first sync

Education

BS Software Engineering · Air University, Islamabad · 2021 – 2025

Languages: English, Urdu, Pashto